



STEM SMART Single Maths 17 - Calculus Revision >

Differentiation from First Principles 2

Pre-Uni Maths for Sciences J3.3 & J3.4

Subject & topics: Maths | Calculus | Differentiation **Stage & difficulty:** A Level P2

Differentiating a function $f(x)$ from first principles involves taking a limit. The derivative of $f(x)$ is given by the expression

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}.$$

Part A

Differentiate $4x^2 + 2x + 7$ from first principles

Differentiate $f(x) = 4x^2 + 2x + 7$ from first principles. Drag and drop options into the spaces below.

$f(x + h) = 4(x + h)^2 + 2(x + h) + 7$. Substituting this into the expression for $f'(x)$,

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{(4(x + h)^2 + 2(x + h) + 7) - (4x^2 + 2x + 7)}{h}.$$

Next, expanding the brackets in the numerator and simplifying gives

$$f'(x) = \lim_{h \rightarrow 0} \frac{\boxed{} - (4x^2 + 2x + 7)}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} (\boxed{} + (\boxed{})h).$$

Finally, take the limit. As $h \rightarrow 0$, the terms containing h tend to 0. Therefore,

$$f'(x) = \boxed{}.$$

Items:

- $8x + 2$
- $8x + 4$
- $4x^2 + 2x + 7 + 4hx + 2h + 4h^2$
- 4
- $4x^2 + 2x + 7 + 8hx + 2h + 4h^2$
- 7
- $4x^2 + 4h^2$

Part B

Differentiate $ax^2 + bx + c$ from first principles

Differentiate $f(x) = ax^2 + bx + c$, where a , b and c are constants, from first principles.

$f(x + h) = a(x + h)^2 + b(x + h) + c$. Substituting this into the expression for $f'(x)$,

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{(a(x + h)^2 + b(x + h) + c) - (ax^2 + bx + c)}{h}.$$

Next, expanding the brackets in the numerator and simplifying gives

$$f'(x) = \lim_{h \rightarrow 0} \frac{\boxed{} + (\boxed{})h + (\boxed{})h^2}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} (\boxed{} + (\boxed{})h).$$

Finally, take the limit. As $h \rightarrow 0$, the terms containing h tend to 0. Therefore,

$$f'(x) = \boxed{}.$$

Items:

$b + ah$

a

$ax^2 + 2ahx + ah^2$

1

$2ax + b$

$2a$

ab

0



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Differentiation (powers of x) 1i

Subject & topics: Maths **Stage & difficulty:** A Level P2

It is given that $f(x) = \frac{1}{x} - \sqrt{x} + 3$.

Part A

Find $f'(x)$

Find $f'(x)$.

The following symbols may be useful: x

Part B

Find $f''(4)$

Find $f''(4)$.

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Integration (powers of x) 2ii

Subject & topics: Maths

Stage & difficulty: A Level P2

Part A

Find integral

Find $\int x(x^2 - 4) \, dx$.

The following symbols may be useful: C, c, k, x

Part B

Evaluate integral

Evaluate $\int_1^6 x(x^2 - 4) \, dx$.

Give the exact value of your answer as a decimal.

Part C

Find integral

Find $\int \frac{6}{x^3} \, dx$.

The following symbols may be useful: C, c, k, x

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Area Under a Curve 2

Pre-Uni Maths for Sciences K2.8

Subject & topics: Maths | Calculus | Integration **Stage & difficulty:** A Level P2

A graph of the functions $y = x^2 + 3$ and $y = 4x$ is shown in **Figure 1**. Find the area of the shaded region labelled A, the region between the line $y = 4x$ and the curve $y = x^2 + 3$.

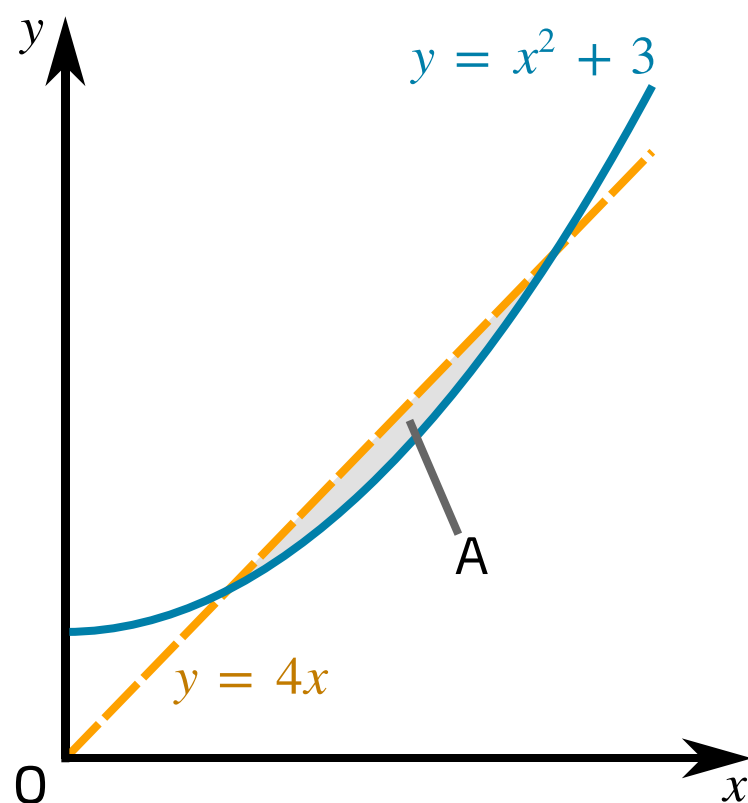


Figure 1: A graph of the functions $y = x^2 + 3$ and $y = 4x$. The shaded region A is the region between the line $y = 4x$ and the curve $y = x^2 + 3$.

Find the area of the region A. Give your answer in the form of an improper fraction.

Created for isaacphysics.org by Julia Riley

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Functions from Differential Equations 2i

Subject & topics: Maths **Stage & difficulty:** A Level P2

The gradient of a curve is given by $\frac{dy}{dx} = 3x^2 + a$, where a is a constant. The curve passes through the points $(-1, 2)$ and $(2, 17)$. Find the equation of the curve.

The following symbols may be useful: x , y

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Calculus

Subject & topics: Maths Stage & difficulty: A Level C2

Part A

Integrating a factorised expression

Find $\int (x^2 + 9)(x - 4) \, dx$.

The following symbols may be useful: C, c, k, x

Part B

Differentiation

A curve has the equation $y = \frac{1}{3}x^3 - 9x$.

Find $\frac{dy}{dx}$.

The following symbols may be useful: Derivative(y, x), x, y

Part C

Stationary points

Find the coordinates of the stationary points of the curve $y = \frac{1}{3}x^3 - 9x$. Enter the x - and y -coordinates of the stationary point with the largest x -coordinate.

(,)

Part D

Nature of stationary point

Determine the nature of the stationary point with the largest x -coordinate.

- ☐ Minimum
- ☐ Maximum
- ☐ Neither/Inconclusive

Part E

Tangent to the curve

Given that $24x + 3y + 2 = 0$ is the equation of the tangent to the curve $y = \frac{1}{3}x^3 - 9x$ at the point (p, q) , find the values of p and q .

(i) Enter value of p :

The following symbols may be useful: p

(ii) Enter value of q :

The following symbols may be useful: q

Part F

Normal to the curve

Find the equation of the normal to the curve $y = \frac{1}{3}x^3 - 9x$ at the point (p, q) you found in Part E.

Give your answer in the form $ax + by + c = 0$, where a , b , and c are integers.

The following symbols may be useful: x, y

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Minimisation: Surface Area

Subject & topics: Maths | Calculus | Differentiation **Stage & difficulty:** A Level P2

Figure 1 shows a solid shape, which is made out of a cuboid and two half-cylinders.

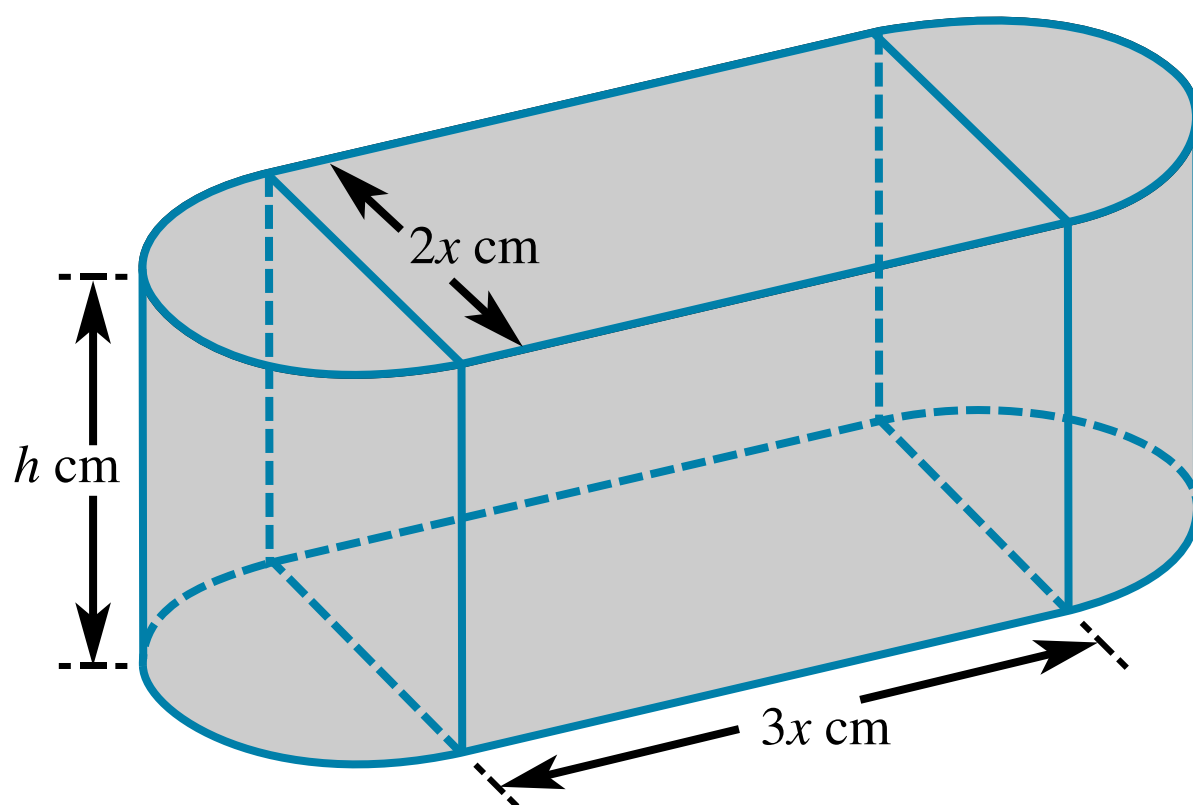


Figure 1: A solid shape made out of a cuboid and two half-cylinders.

Part A

Expression for surface area

Find an expression for the surface area of the shape S in terms of π , x and h .

The following symbols may be useful: S , h , π , x

Part B

Expression for volume

Find an expression for the volume of the shape V in terms of π , x and h .

The following symbols may be useful: v , h , π , x

Part C

Minimum surface area

If the volume of the shape is $32\,000\text{ cm}^3$, find the value of x for which the surface area is a minimum. Give your answer to 3 significant figures.

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